

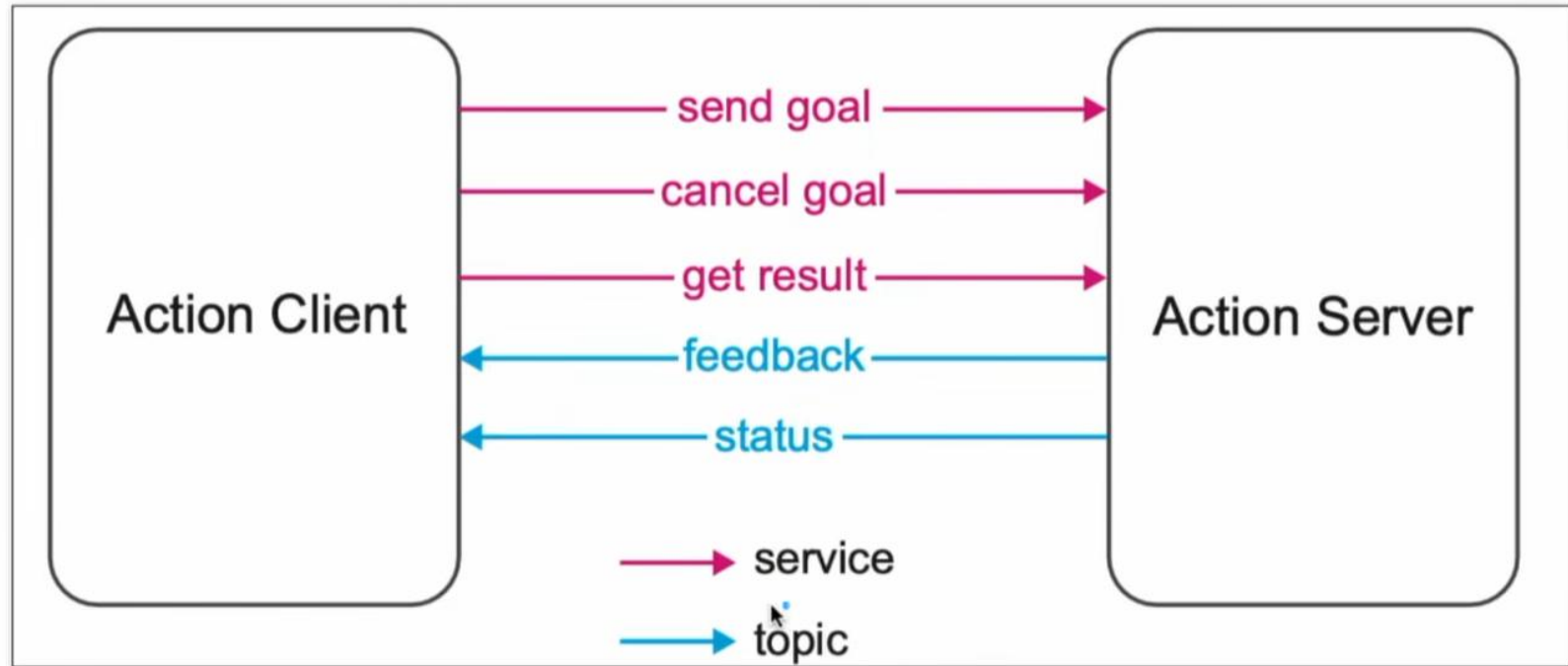
세미나 5회차: Action 구현

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1. Action

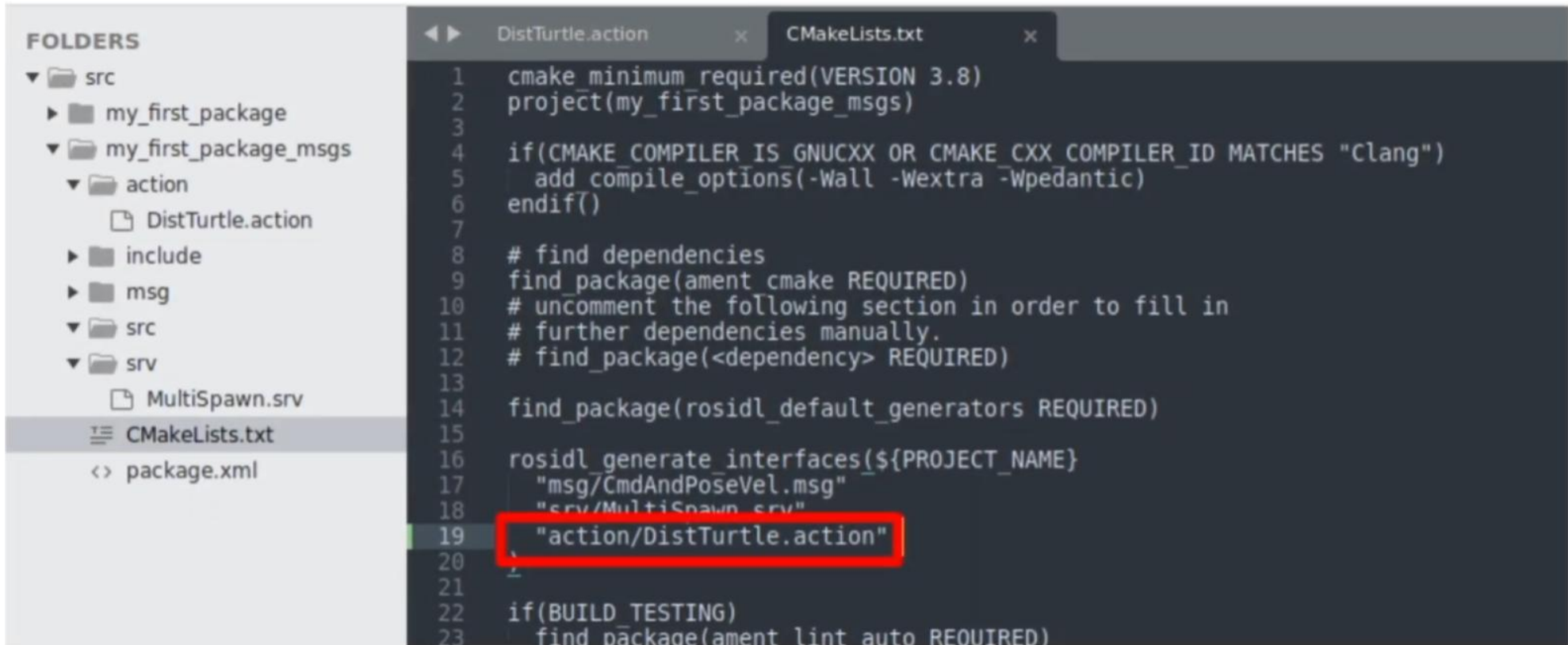


디렉토리 생성

```
lina@lina-ubuntu:~/ros2_study/src/my_first_package_msgs$  
lina@lina-ubuntu:~/ros2_study/src/my_first_package_msgs$ mkdir action  
lina@lina-ubuntu:~/ros2_study/src/my_first_package_msgs$ ls  
action CMakeLists.txt include msg package.xml src srv  
lina@lina-ubuntu:~/ros2_study/src/my_first_package_msgs$
```

DistTurtle.action이라는 액션 정의 파일 생성

```
action > ≡ DistTurtle.action  
1  #Request  
2  float32 linear_x  
3  float32 angular_z  
4  float32 dist  
5  ---  
6  #Result  
7  float32 pos_x  
8  float32 pos_y  
9  float32 pos_theta  
10 float32 result_dist  
11 ---  
12 #Feedback  
13 float32 remained_dist
```



CMakeLists.txt: 빌드 시스템 도구인 CMake에게 이 프로젝트를 어떻게 빌드(컴파일 및 패키징)해야 하는지 알려주는 설명서(action, msg 등 폴더 안의 파일들을 ROS2가 인식할 수 있는 파이썬/C++ 코드로 자동 변환하도록 설정하는 명령어 포함)

이 파일은 "내 패키지 폴더 안에서 파일이 새로 추가되거나 구조가 바뀔 때" 수정합니다.

```
DistTurtle.action x CMakeLists.txt x package.xml x
1 <?xml version="1.0"?>
2 <?xml-model href="http://download.ros.org/schema/package_format3.xsd"
  schematypens="http://www.w3.org/2001/XMLSchema"?>
3 <package format="3">
4   <name>my first package msgs</name>
5   <version>0.0.0</version>
6   <description>TODO: Package description</description>
7   <maintainer email="pw@todo.todo">pw</maintainer>
8   <license>TODO: License declaration</license>
9
10  <buildtool_depend>ament_cmake</buildtool_depend>
11
12  <test_depend>ament_lint_auto</test_depend>
13  <test_depend>ament_lint_common</test_depend>
14
15  <build_depend>roscpp default generators</build_depend>
16  <exec_depend>roscpp default runtime</exec_depend>
17  <member_of_group>roscpp_interface_packages</member_of_group>
18
19  <export>
20    <build_type>ament_cmake</build_type>
21  </export>
22 </package>
23
```

package.xml: 이 ROS2 패키지가 어떤 정보를 담고 있고, 실행하려면 어떤 다른 패키지들이 필요한지 적어두는 파일

이 파일은 "외부에서 새로운 패키지를 가져와야 할 때" 수정합니다.

Action 정의 성공

admin2@admin2: ~/ros2_study

admin2@admin2: ~/ros2_study 150x24

```
admin2@admin2:~/ros2_study$ colcon build
Starting >>> my_first_package
Starting >>> my_first_package_msgs
Finished <<< my_first_package [1.98s]
Finished <<< my_first_package_msgs [3.70s]

Summary: 2 packages finished [4.29s]
admin2@admin2:~/ros2_study$ source install/setup.bash
admin2@admin2:~/ros2_study$ ros2 interface show my_first_package_msgs/action/DistTurtle
#Request
float32 linear_x
float32 angular_z
float32 dist
---
#Result
float32 pos_x
float32 pos_y
float32 pos_theta
float32 result_dist
---
#Feedback
float32 remained_dist
admin2@admin2:~/ros2_study$
```

Action 서버 생성

```
dist_turtle_action_server.py
ros2_study > src > my_first_package > my_first_package > dist_turtle_action_server.py
1  import rclpy as rp
2  from rclpy.node import Node
3  from rclpy.action import ActionServer
4  from my_first_package_msgs.action import DistTurtle
5  from turtlesim.msg import Pose
6
7  class DistTurtleServer(Node):
8
9      def __init__(self):
10         super().__init__('dist_turtle_action_server')
11         self._action_server = ActionServer(
12             self,
13             DistTurtle,
14             'dist_turtle',
15             self.execute_callback
16         )
17
18     def execute_callback(self, goal_handle):
19         goal_handle.succeed()
20         result = DistTurtle.Result()
21         return result
22
23     def main(args=None):
24         rp.init(args=args)
25         dist_turtle_action_server = DistTurtleServer()
26         rp.spin(dist_turtle_action_server)
27         dist_turtle_action_server.destroy_node()
28         rp.shutdown()
29
30     if __name__ == '__main__':
31         main()
```

Setup.py 수정

```
entry_points={
    'console_scripts': [
        'my_first_node = my_first_package.my_first_node:main',
        'dist_turtle_action_server = my_first_package.dist_turtle_action_server:main'
    ],
},
```

- [setup.py](#): 파이썬 기반의 ROS2 패키지가 어떻게 설치되고 시스템에 등록되어야 하는지 빌드 도구에게 알려주는 파이썬 패키지 전용 배포/설치 설명서
- ROS2에서 '나만의 메시지(msg), 서비스(srv), 액션(action)'을 만드는 패키지는 무조건 CMake 기반으로만 만들어야 한다 → 그래서 아까 CMake 파일을 수정했다.

동작 확인

```
admin2@admin2: ~/ros2_study

float32 pos_theta
float32 result_dist
---
#Feedback
float32 remained_dist
admin2@admin2:~/ros2_study$ code
admin2@admin2:~/ros2_study$ code ..
admin2@admin2:~/ros2_study$ colcon build
Starting >>> my_first_package
Starting >>> my_first_package_msgs
Finished <<< my_first_package_msgs [0.55s]

Finished <<< my_first_package [0.73s]

Summary: 2 packages finished [0.93s]
admin2@admin2:~/ros2_study$ so
soelim                software-properties-gtk  sort
source
soffice                sol                      sotruss

admin2@admin2:~/ros2_study$ source install/setup.bash
admin2@admin2:~/ros2_study$ ros2 run my_first_package dist_turtle_action_s
erver

admin2@admin2:~/ros2_study$

admin2@admin2:~/ros2_study$

Starting >>> my_first_package_msgs
Finished <<< my_first_package_msgs [0.57s]
Finished <<< my_first_package [0.73s]

Summary: 2 packages finished [0.92s]
admin2@admin2:~/ros2_study$ source install/setup.bash
admin2@admin2:~/ros2_study$ ros2 action send_goal /dist_turtle my_first_p
ackage_msgs/action/DistTurtle "{linear_x: 0, angular_z: 0, dist: 0}"
Waiting for an action server to become available...
Sending goal:
    linear_x: 0.0
    angular_z: 0.0
    dist: 0.0

Goal accepted with ID: 6d9941ef433042cd9a55d76e206fa8e7

Result:
    pos_x: 0.0
    pos_y: 0.0
    pos_theta: 0.0
    result_dist: 0.0

Goal finished with status: SUCCEEDED
admin2@admin2:~/ros2_study$
```

Feedback 구현

```
dist_turtle_action_server.py x DistTurtle.action x setup.py x
1 import rclpy as rp
2 from rclpy.action import ActionServer
3 from rclpy.node import Node
4 → import time
5
6 from my_first_package_msgs.action import DistTurtle
7
8
9 class DistTurtleServer(Node):
10
11     def __init__(self):
12         super().__init__('dist_turtle_action_server')
13         self.action_server = ActionServer(
14             self,
15             DistTurtle,
16             'dist_turtle',
17             self.execute_callback)
18
19     def execute_callback(self, goal_handle):
20         feedback_msg = DistTurtle.Feedback()
21         for n in range(0,10):
22             feedback_msg.remained_dist = float(n)
23             goal_handle.publish_feedback(feedback_msg)
24             time.sleep(0.5)
25
26         goal_handle.succeed()
27         result = DistTurtle.Result()
28         return result
29
```

callback 함수 부분에 feedback 구문을 추가하고 이후 send_goal 명령어를 실행할 때 - -feedback 옵션을 추가하면 feedback이 실행된다.

```
admin2@admin2: ~/ros2_study
self._spin_once_impl(timeout_sec)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors
.py", line 797, in _spin_once_impl
    handler, entity, node = self.wait_for_ready_callbacks(timeout_sec=time
out_sec)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors
.py", line 780, in wait_for_ready_callbacks
    return next(self._cb_iter)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors
.py", line 677, in _wait_for_ready_callbacks
    wait_set.wait(timeout_nsec)
KeyboardInterrupt
[ros2run]: Interrupt
admin2@admin2:~/ros2_study$ colcon build
Starting >>> my_first_package
Starting >>> my_first_package_msgs
Finished <<< my_first_package_msgs [0.55s]
Finished <<< my_first_package [0.73s]

Summary: 2 packages finished [0.92s]
admin2@admin2:~/ros2_study$ source install/setup.bash
admin2@admin2:~/ros2_study$ ros2 run my_first_package dist_turtle_action_s
erver

admin2@admin2: ~/ros2_study
Summary: 2 packages finished [0.91s]
admin2@admin2:~/ros2_study$ source install/setup.bash
admin2@admin2:~/ros2_study$ ros2 action send_goal --feedback /dist_turtle
my_first_package_msgs/action/DistTurtle "{linear_x: 0, angular_z: 0, dis
t: 0}"
Waiting for an action server to become available...
Sending goal:
    linear_x: 0.0
    angular_z: 0.0
    dist: 0.0
Goal accepted with ID: 171e3ee6ec7248e7961efa7e4aa6402a

Feedback:
    remained_dist: 0.0

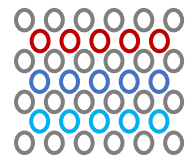
Feedback:
    remained_dist: 1.0

Feedback:
    remained_dist: 2.0

Feedback:
    remained_dist: 3.0
```

Q & A

Thank you for your attention



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